



Case Study 1: Precision Urban Agriculture for Vertical Farming

SeedDown

Drop a Seed Take a Seat

Smart IoT + AI + Community
Platform for Urban Farmers

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Why do urban farmers give up?

73%

of new urban farmers
quit within 6 months



No Knowledge

No farming background or training



No Guidance

Nobody to ask when things go wrong



High Failure Cost

Energy bills + dead crops = losses



No Data

Can't monitor or optimise their farm

Vertical farming boom is collapsing

60%

of revenue

Energy Crisis

LED + HVAC costs eat
40–60% of all revenue

3×

more expensive

No Price Advantage

Cannot compete with
traditional agriculture

80%

startups failed

Wrong Model

Scaled like SaaS,
but operated like a factory

One platform. Two modes. Total control.

SeedDown is an AI + IoT smart vertical farming platform connecting ESP32 sensors, Node.js backend, Firebase Firestore, and a responsive web dashboard.

1



Collect

ESP32 reads Temp, Humidity, Gas, Moisture, pH, Light, Water distance

2



Store

Data sent to Node.js backend and saved into Firebase Firestore in real time

3



Analyse

AI checks readings against farm thresholds and detects abnormal conditions

4



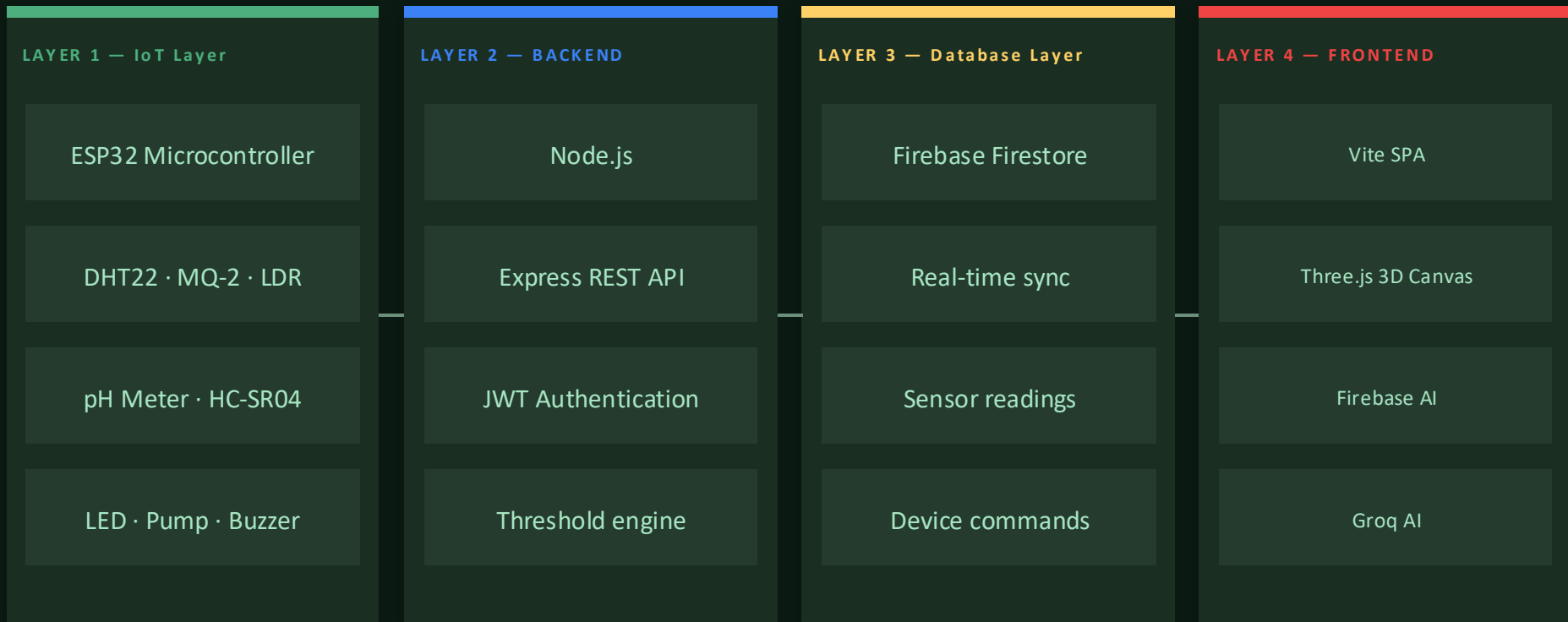
Act

System returns the right command to ESP32 — pump, light, buzzer, or alert

Special: Real IoT Loop; AI works with deterministic fallback rules; two complete modes

Data Flow: IoT Sensors → Firebase → Node.js API → AI Claude → Frontend SPA → User

Scalable, modular, cloud-native stack



Data Flow: IoT Sensors → Firebase → Node.js API → AI Claude → Frontend SPA → User



DEMO





SeedDown

SMART FARM MANAGEMENT

SELECT FARMING MODE



Beginner

Guided experience



Commercial

Pro data dashboard

Sign In

Register

Email

wongjiahui@gmail.com

Password

•••••



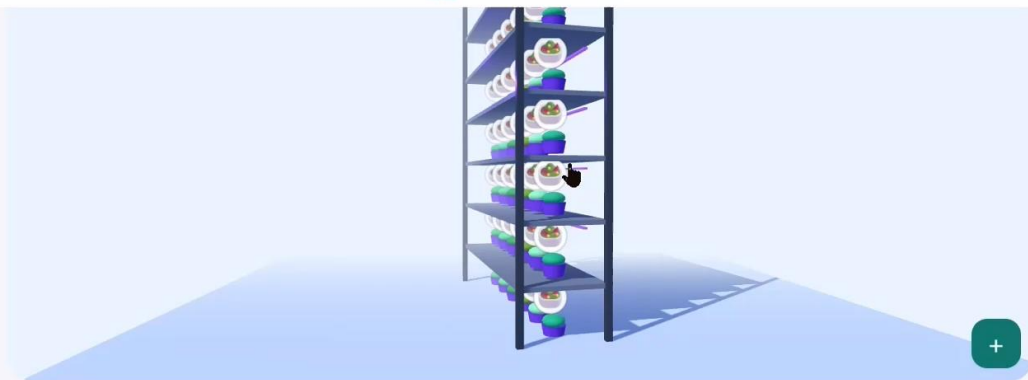
[Forgot password?](#)

Sign In →

or

 View as Guest (Demo)

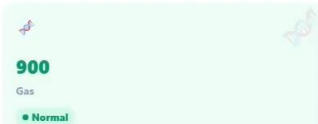
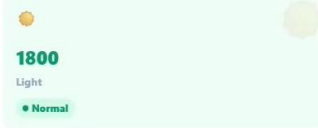
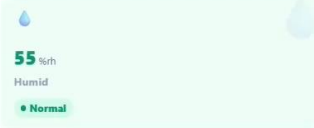
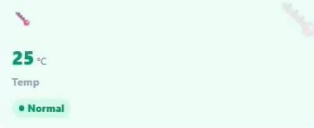
Powered by next level .idm



Live Data

beginner_standard - 7bw9nzeCOhX3Z3Ku9xs - standa...

LIVE



COMMERCIAL DIGITAL TWIN
Botani
3-Zone Commercial Farm · 36/36 planted

100% occupied

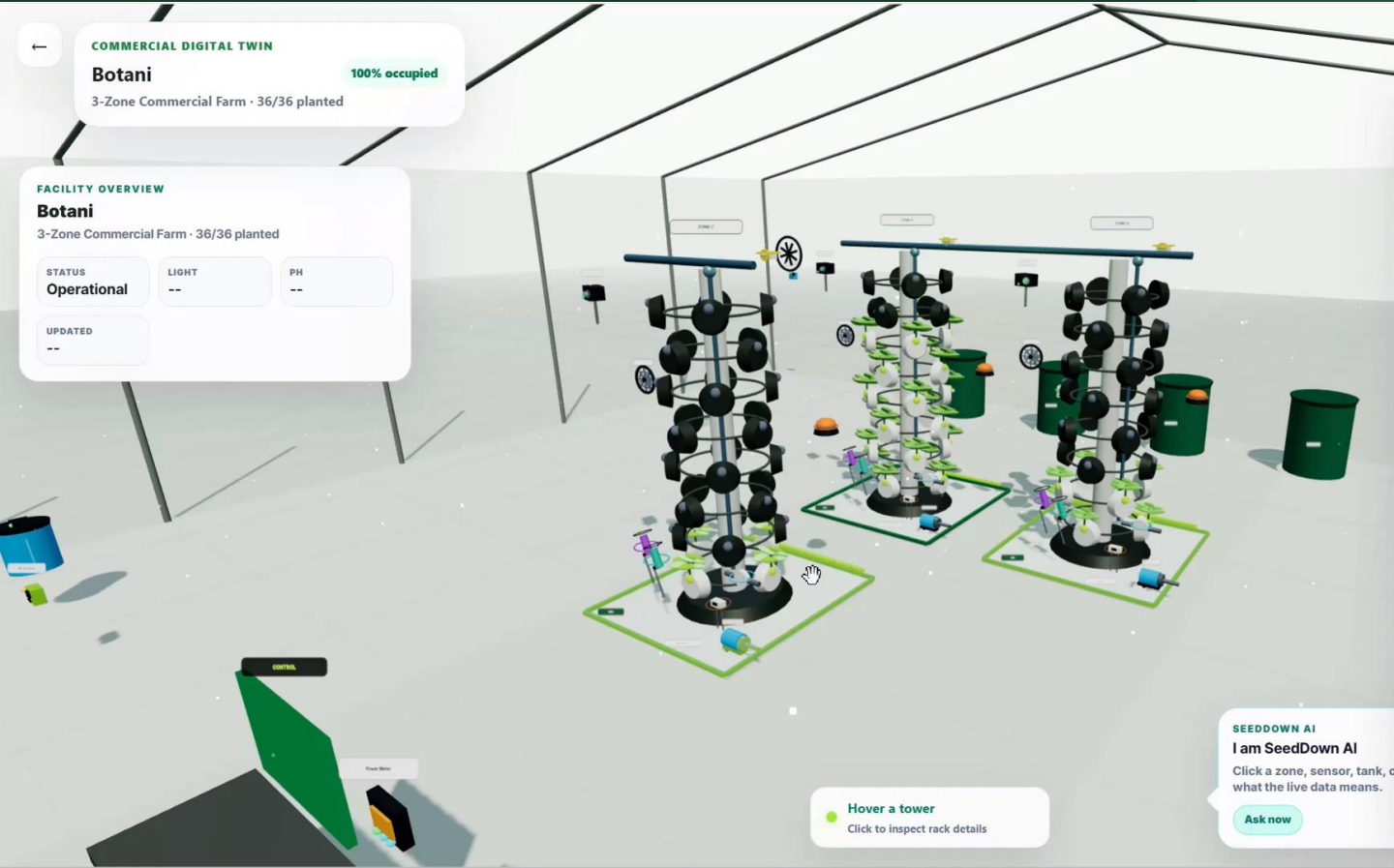
FACILITY OVERVIEW
Botani
3-Zone Commercial Farm · 36/36 planted

STATUS
Operational

LIGHT
--

PH
--

UPDATED
--



HIDE PANEL

OPERATIONS
Farm Command Center

ZONE A
Lettuce
24/12 slots · commercial_zone_node_01001

31°C · 65% · pH -- · Last updated 12:46 AM
-- Tap to drill into zone

ZONE B
Basil
8/12 slots · commercial_zone_node_01002

26°C · 60% · pH 6.3 · Last updated 12:46 AM
-- Tap to drill into zone

ZONE C
Marigold
4/12 slots · commercial_zone_node_01003

23°C · 63% · pH 6 · Last updated 12:46 AM
-- Tap to drill into zone

TOOLS

What-if

Control

Disease

Camera

ECG

Alerts

Who we serve — and why they'll pay



Students & Hobbyists

UTM campus & urban communities wanting to grow food at home or in shared spaces

Free tier + coin economy + Ads



Commercial Farms

SME vertical farms needing profit optimisation, ESG reporting & automation

SaaS subscription RM 99–299/mo

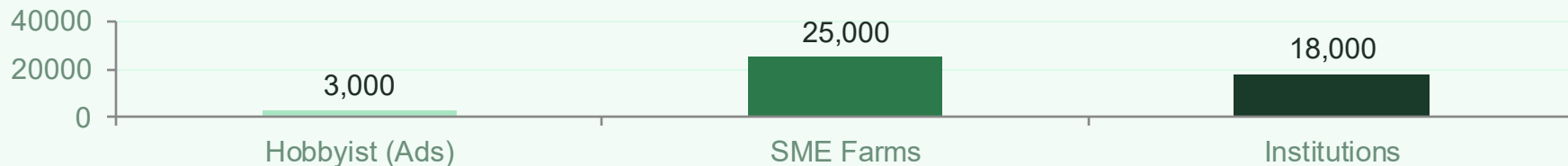


Institutions

Universities, schools & government bodies running food security or sustainability programs

Enterprise licensing

Projected Monthly Revenue Potential (RM)



Affordable hardware — deployable from RM 30

RM 30–60

Beginner Starter

DHT, soil, light

RM 80–150

Beginner Standard

+ pH, gas, water

RM 170–330

Beginner Pro

+ EC, CO₂**RM 250–650**

Commercial Zone

Full zone sensors

RM 130–320

Commercial Master

Farm-level ctrl

Component	Function	Approx. Cost (RM)
NodeMCU ESP32	Main controller (WiFi + GPIO)	11 – 25
DHT11 / DHT22	Air temperature + humidity	6 – 13
Soil moisture sensor	Root-zone moisture trend	3 – 6
Analog pH sensor kit	Nutrient acidity monitoring	25 – 60
EC / TDS sensor	Nutrient strength estimate	25 – 90
Pump + LED + Fan + Relay	Actuators (water, light, vent)	12 – 40

SeedDown RoadMap

1.

- Hackathon Validation & ESP32 pilot kits

2.

- Urban Farming Communities
- Schools
- Makerspaces
- Hydroponic Retailers

3.

- Commercial Pilots with stronger camera AI, forecasting, ESG reporting and hardware reliability



Drop A Seed. Take A Seat.

SeedDown turns any urban farmer into a smart, data-driven grower
— from a university student to a commercial vertical farm operator.

IoT Automation

AI Intelligence

Community Economy